

Binomial expansion - Set 2 - Solutions

Specification MM2.4.

Answer key

Question	Answer	Topic
1	H	Binomial expansion
2	C	Binomial expansion
3	H	Binomial expansion
4	B	Binomial expansion

Worked solutions

Question 1 BIN-12 • Binomial expansion • spec MM2.4

The expansion of $\left(2x - \frac{1}{x^2}\right)^9$ contains a term in x^k for each of which of the following values of k ?

I $k = 0$

II $k = 3$

III $k = -6$

Answer: H

General term: $\binom{9}{r}(2x)^{9-r}(-x^{-2})^r$ has power $9 - r - 2r = 9 - 3r$, which takes values 9, 6, 3, 0, -3, ..., -18. $k = 0$ needs $r = 3$: yes. $k = 3$ needs $r = 2$: yes. $k = -6$ needs $r = 5$: yes. All three.

Question 2 BIN-15 • Binomial expansion • spec MM2.4

The coefficient of x^2 in $(1+x)^n + (1+x)^{n+1}$ is 36. Find n .

Answer: C

$$\binom{n}{2} + \binom{n+1}{2} = \frac{n(n-1) + (n+1)n}{2} = n^2 = 36, \text{ so } n = 6.$$

Question 3 X2-13 • Binomial expansion • spec MM2.4

n is a positive integer. Which of the following statements is/are true?

I The coefficient of x^n in $(1+x)^{2n}$ is $\binom{2n}{n}$.

II The sum of the coefficients in the expansion of $(1-x)^n$ is 0.

III $(1+x)^n \geq 1 + nx$ for all $x \geq 0$.

Answer: H

I is the general term with $r = n$. II: put $x = 1$. III: expanding $(1+x)^n$ gives $1 + nx$ plus non-negative terms when $x \geq 0$.

Question 4 BIN-14 • Binomial expansion • spec MM2.4

How many terms in the expansion of $(x+y+z)^4$ (after collecting like terms) are there?

Answer: B

Terms are $x^a y^b z^c$ with $a+b+c=4$, $a, b, c \geq 0$. Counting: for $a=0$ there are 5 choices of (b, c) , $a=1$: 4, ..., $a=4$: 1. Total $5 + 4 + 3 + 2 + 1 = 15$.